Your Spending is Your Fingerprint: Behavioral Clustering of Credit Card Transactions

Elliot Self & Angelina Schapiro

Abstract

Prior work on human mobility and burstiness metrics [1, 2] suggests behavioral signals encode identity. We model consumer behavior as a behavioral fingerprint across three dimensions: spending rhythm, geographic range, and category diversity, derived from 10 years of credit card transactions. Applying K-Means clustering to 500 consumers, we identify four consumer segments: Engaged Explorers, Casual Locals, Routine Specialists, and Erratic Minimalists.

Research Questions

Can spending behavior alone reveal what consumer segment you belong to?

- **Which dimension—rhythm, range, or variety—best separates consumers into distinct segments?**

Dataset Description

We used the IBM Synthetic Credit Card Transaction Dataset, a collection of simulated consumer transactions designed to mirror real spending behavior. It includes 6,164,604 transactions across 2,000 users spanning 2000 to 2020. We randomly sampled 500 users and restricted our analysis to 2009 to 2019, the ten-year window with the most complete transaction coverage.

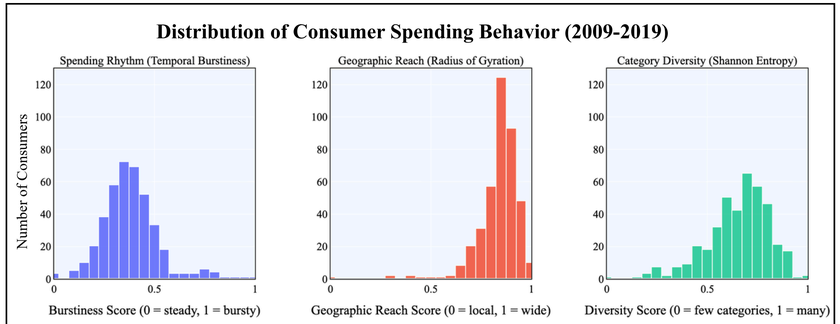
To enrich the raw transaction data, we performed the following steps:

- **MCC Enrichment: queried a live MCC code repository via REST API and applied a custom keyword taxonomy to collapse ~650 raw descriptions into 12 interpretable behavior categories such as Food & Dining, Travel, and Groceries.**
- **Geocoding: resolved each unique zip code to a latitude/longitude coordinate using the pgeocode Nominatim library, enabling our spatial displacement metric.**

Behavioral Metrics

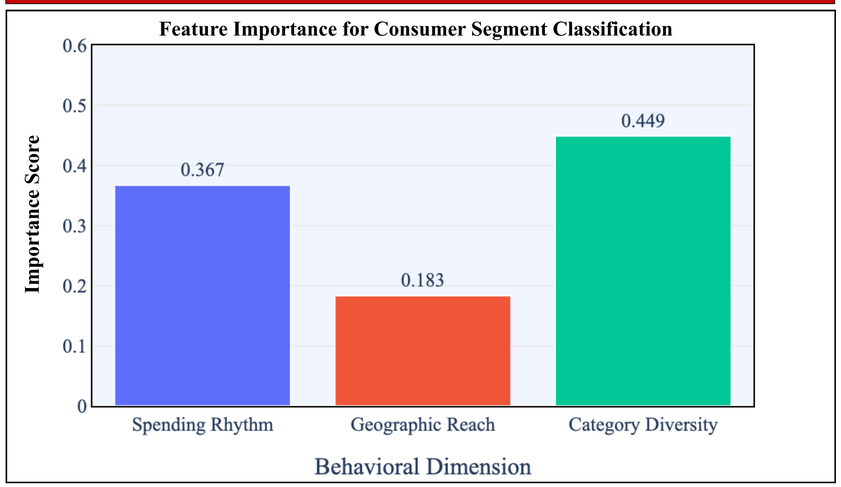
Metric	Description	Formula
Temporal Burstiness (B) (Rhythm)	Measures spending irregularity using time gaps between consecutive transactions, averaged across 2009–2019	$B = \frac{\sigma - \mu}{\sigma + \mu}$
Spatial Displacement (Rg) (Geographic Range)	Haversine distance from each transaction to the user's most frequent zip code (home anchor).	$R_g = \sqrt{\frac{1}{N}\sum_{i=1}^{N} d_i^2}$
Categorical Shannon entropy over MCC spending categories (H) (Category Diversity)	Shannon entropy over MCC spending categories	$H_{norm} = \frac{H}{\log_2(K)}$

Data Exploration



Spending rhythm and category diversity spread across the full 0 to 1 range after normalization, while geographic reach clusters near 0.9, reflecting that most consumers travel widely over 10 years. This concentration makes geographic reach the weakest clustering feature. All three dimensions were needed to identify meaningful consumer segments.

Feature Analysis

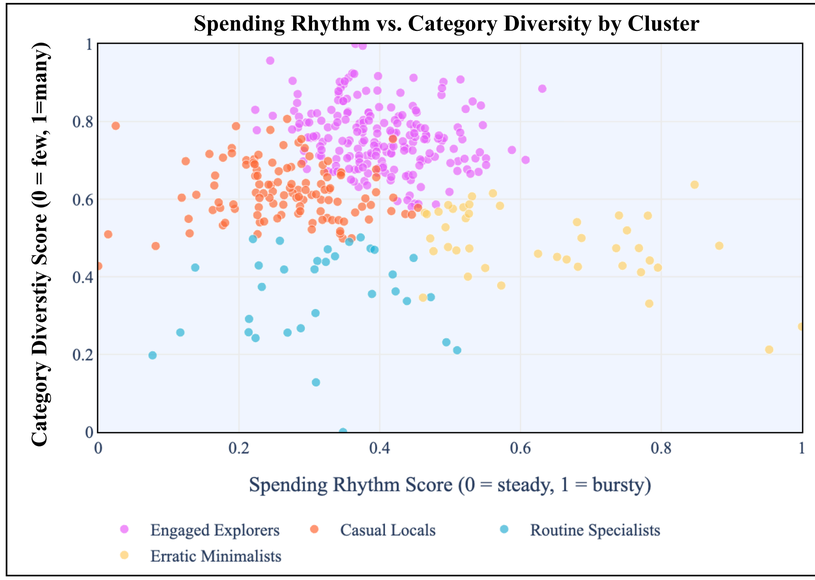


A Random Forest Classifier confirms Category Diversity as the strongest predictor of cluster membership (0.449), followed by Spending Rhythm (0.367) and Geographic Reach (0.183), consistent with its narrow distribution seen above.

Data Analysis

Although K-Means clustered users across all three normalized metrics, Category Diversity and Spending Rhythm were the strongest predictors, so we visualize these two dimensions below. K=4 was selected to balance accuracy and interpretability:

- **Engaged Explorers: many categories, moderate rhythm, travel widely**
- **Casual Locals: moderate categories, steady rhythm, stay local**
- **Routine Specialists: few categories, moderate rhythm, travel widely**
- **Erratic Minimalists: few categories, bursty rhythm, stay local**



Consumer Segment Profiles

Segment	Spending Rhythm	Geographic Reach	Category Diversity	% of Users
Engaged Explorers	0.4	0.88	0.76	52.1%
Casual Locals	0.27	0.79	0.62	29.2%
Routine Specialists	0.32	0.88	0.35	8.2%
Erratic Minimalists	0.63	0.79	0.49	10.5%

Over half of consumers are Engaged Explorers. Wide-ranging spending is the norm, which explains why geographic reach is the weakest distinguishing feature

Conclusion

Spending behavior alone can meaningfully distinguish consumer segments. K-Means identifies four consumer segments, with Random Forest classification confirming Category Diversity and Spending Rhythm as the strongest separating dimensions. These behavioral metrics could serve as powerful input features to improve existing ML fraud models. A sudden drop in category entropy (spending concentrated in one category) or spike in burstiness over a short window (many transactions clustered in a short period) are signals that existing systems could incorporate. Future work could compute these metrics on rolling time windows for real-time detection.

[1] Goh, K.I. & Barabási, A.L. (2008). Burstiness and memory in complex systems. EPL, 81(4), 48002.
[2] Gonzalez, M.C., Hidalgo, C.A. & Barabási, A.L. (2008). Understanding individual human mobility patterns. Nature, 453, 779–782.